📷 200W USB Camera Introduction and Usage

1. Course Overview

This course will introduce the basic characteristics of the 200W pixel USB HD wide-angle camera equipped on the A1 Pro robot, and demonstrate how to use the ROS 2 driver to start the camera node, use `rqt_image_view` to view the color image stream in real-time, and read the camera intrinsic parameters `/camera_info` topic.

🎯 Course Objectives

1. Understand the hardware specifications and operating parameters of the 200W USB camera
2. Master how to start the camera driver and image visualization using `usb_cam`
3. Master how to view and understand the camera intrinsic topic `/camera_info`

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2. Camera Hardware Introduction



Lens plan	Cmosm
Image Sensor	1/2.7inch OV2710
Lens	F2.8mm
Picture pixels	2MP
Image format	YUV/MJPG
Resolution	240p/480p/720p/1080p
Max frame rate	120fps
FOV (diagonal)	148°
Distortion	-17%
USB	USB2.0 No need drive
Camera size	32*32*29.5mm

The USB HD camera equipped on A1 Pro features wide dynamic range, high frame rate, and plug-and-play capability:

- **Sensor Resolution**: Up to 1920×1080 (1080P) / Calibration default output 640×480
- **Working Frame Rate**: 30 FPS
- **Field of View (FOV)**: Large field of view with no distortion/minor distortion wide-angle lens
- **Interface Type**: USB 2.0 driver-free UVC standard protocol
- **Data Output Format**: YUYV / MJPEG

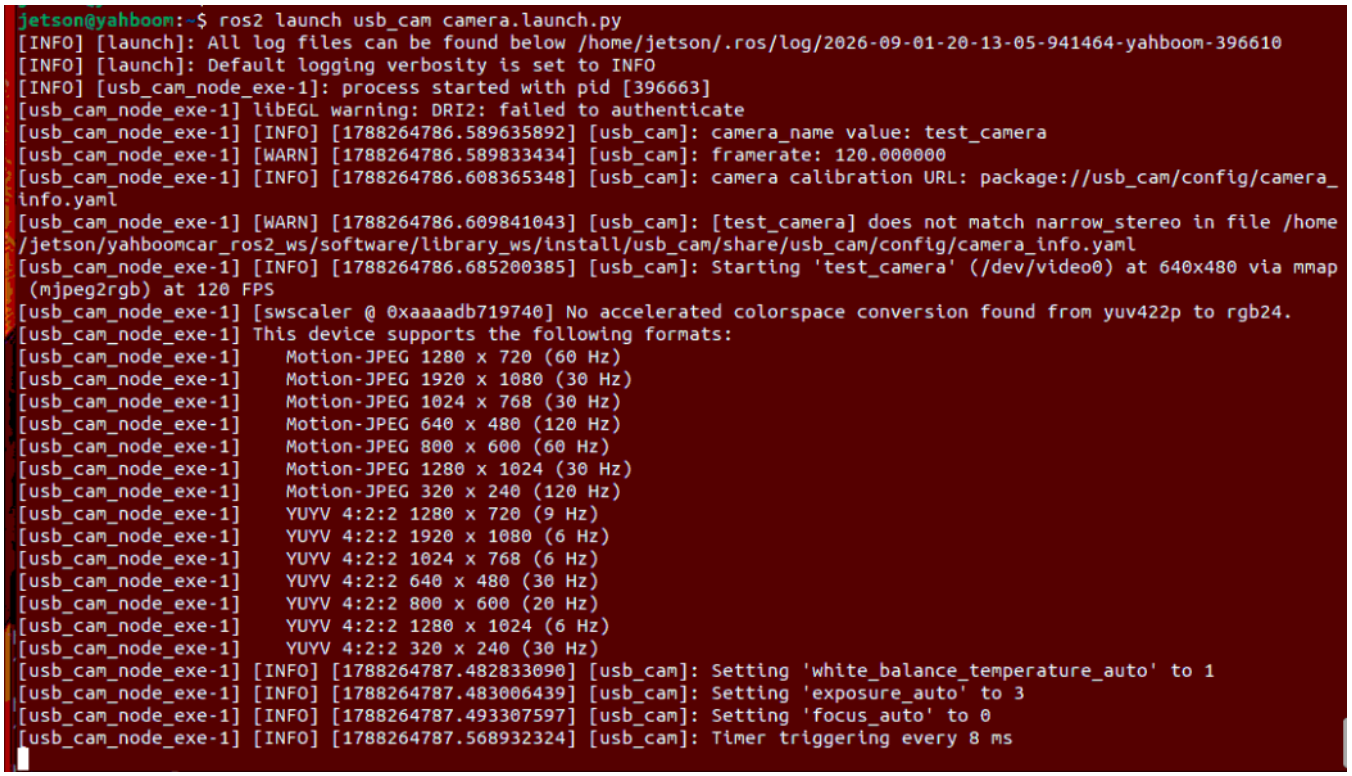
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3. Preparation and Driver Startup

3.1 Start USB Camera Driver Node

Open the terminal and run the following command to start the camera driver:

```
ros2 launch usb_cam camera.launch.py
```



After the driver starts successfully, it will continuously capture images in the background and publish `/image_raw` (or `/usb_cam/image_raw`) and `/camera_info` topics.

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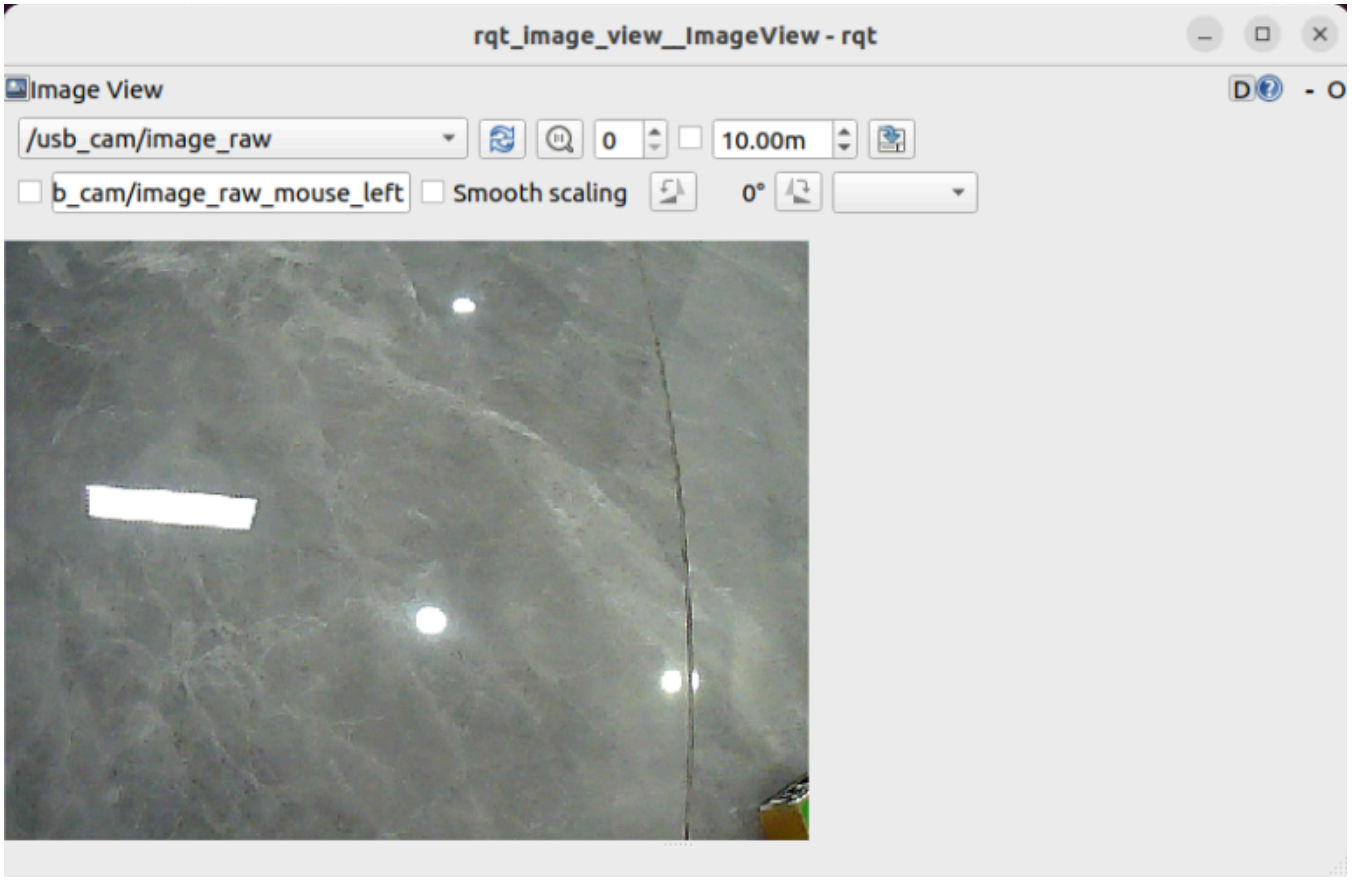
4. Hands-on Demo and Visualization

4.1 Use rqt_image_view to View Real-time Images

Run the following in a new terminal:

```
ros2 run rqt_image_view rqt_image_view
```

In the pop-up interface, select the camera's image topic (`/usb_cam/image_raw`) from the dropdown list on the upper left to view the RGB color image captured by the camera in real-time:



4.2 View Camera Intrinsic Topic (`/usb_cam/camera_info`)

Execute in the terminal:

```
ros2 topic echo /usb_cam/camera_info --once
```

```
jetson@yahboom:~$ ros2 topic echo /usb_cam/camera_info --once
header:
  stamp:
    sec: 1788264870
    nanosec: 618827000
  frame_id: camera
height: 480
width: 640
distortion_model: plumb_bob
d:
- -0.44136
- 0.203085
- -0.001104
- -0.005037
- 0.0
k:
- 558.96443
- 0.0
- 340.84835
- 0.0
- 559.59295
- 241.81847
- 0.0
- 0.0
- 1.0
r:
- 1.0
- 0.0
- 0.0
- 0.0
- 1.0
- 0.0
- 0.0
- 0.0
- 1.0
p:
- 469.43236
- 0.0
- 343.63664
- 0.0
- 0.0
- 510.68501
- 241.79933
- 0.0
- 0.0
- 0.0
- 1.0
- 0.0
binning_x: 0
binning_y: 0
roi:
  x_offset: 0
```

5. Core Source Code and Parameter Analysis

5.1 Launch File

Path:

`$HOME/yahboomcar_ros2_ws/software/library_ws/src/usb_cam/launch/camera.launch.py`

```python
def generate_launch_description():
    camera_node = Node(
        package='usb_cam',
        executable='usb_cam_node_exe',
        name='usb_cam',
        output='screen',
        parameters=[
            {'video_device': '/dev/video0'},
            {'image_width': 640},
            {'image_height': 480},
            {'framerate': 30.0},
            {'pixel_format': 'yuyv2rgb'},
            {'io_method': 'mmap'},
            {'camera_frame_id': 'camera_link'},
```

```
        ],
    )
    return LaunchDescription([camera_node])
```

5.2 Core Parameter Description

Parameter Name	Default Value	Description
`video_device`	`/dev/video0`	Video device node path
`image_width`	`640`	Image output width (pixels)
`image_height`	`480`	Image output height (pixels)
`framerate`	`30.0`	Camera capture frame rate (FPS)
`pixel_format`	`yuyv2rgb`	Pixel format conversion (converts raw YUYV to RGB output)
`io_method`	`mmap`	Image memory mapping acquisition method (efficient with no extra memory copy)
`camera_frame_id`	`camera_link`	Camera name in ROS TF coordinate system
`autoexposure`	`true`	Whether to enable automatic exposure control
`auto_white_balance`	`true`	Whether to enable automatic white balance